# Physics-Informed Super-Resolution of Two-Dimensional Turbulence with a U-Net

## Reconstruction at Unseen Reynolds Numbers, and the Trade-off Between Pointwise Accuracy and Fine-Scale Statistics

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**Abstract**

A U-Net reconstructs $128 \times 128$ fields of velocity, pressure and vorticity of two-dimensional turbulence from $4\times$ block-averaged ($32 \times 32$) inputs. The training data is direct numerical simulation (DNS) of the incompressible Navier–Stokes equations in a periodic box, for forced (Kolmogorov) and decaying turbulence. The model is tested only on Reynolds numbers it never saw: $Re = 320$, 800 and 2000 (interpolation) and 4000 (extrapolation), from runs with their own random seeds. On this test set the velocity error is 0.9% (forced) and 1.3% (decaying), against 4.4% and 6.1% for bicubic interpolation of the same input. The vorticity error is 8.4% and 10.1%, against 24.1% and 27.0%. The predicted velocity is exactly divergence-free by construction, and the error grows smoothly with $Re$, including the extrapolation to $Re = 4000$. Pointwise losses leave the finest scales about $10\times$ too weak in energy. An optional spectral loss restores their energy but raises the pointwise error, so the default model and a spectral variant are recommended for different purposes.

# 1 Introduction

Turbulent flows contain motion on a wide range of scales, and resolving all of them is expensive. A super-resolution model takes a coarse field, which is cheap to compute or measure, and predicts the missing fine scales. Fukami *et al.* [4] showed that convolutional networks can do this for turbulent flows.

The useful question is not whether a model fits its training data, but whether it works on flows it has not seen. This project therefore

- splits the data by whole Reynolds numbers, so that no run at a test $Re$ is used in training;
- tests at both interpolated and extrapolated $Re$;
- compares the model with three baselines built from the same coarse input, one of which checks that the test runs are independent of the training runs;
- measures both pointwise error and fine-scale energy, because the two respond differently to the choice of loss.

An earlier version of this pipeline handled only lid-driven cavity flow. The cavity remains in the code as a wall-bounded test case (Section 6), but the results here are for turbulence.

# 2 Turbulence Data

## 2.1 Governing equations

The solver integrates the 2D incompressible Navier–Stokes equations in vorticity form on the periodic box $[0, 2\pi)^2$:

$$\frac{\partial \omega}{\partial t} + \mathbf{u} \cdot \nabla \omega = \nu \nabla^2 \omega - \alpha\, \omega + f, \qquad \mathbf{u} = \left( \frac{\partial \psi}{\partial y}, -\frac{\partial \psi}{\partial x} \right), \quad \nabla^2 \psi = -\omega, \tag{1}$$

with $Re = 1/\nu$, since velocities and lengths are of order one.

Table 1: The two turbulence regimes.

| | Forced | Decaying |
|---|---|---|
| Forcing $f$ | $-4\cos(4y)$, the curl of $\sin(4y)\,\hat{x}$ | none |
| Drag $\alpha$ | 0.1 | 0 |
| Initial field | weak random vorticity, $k \le 8$ | random, $E(k) \propto k^4 e^{-2(k/8)^2}$ |
| Snapshots | after $t = 20$, every 2 time units | from $t = 2$, every 0.5 |

The forced case is the set-up of Kochkov *et al.* [1]. It reaches a statistically stationary

state, in which the drag removes the energy that piles up at large scales. In the decaying case random vortices merge and lose energy.

In two dimensions energy cascades to large scales and enstrophy to small ones [2,3]. Above the forcing wavenumber the inviscid theory predicts $E(k) \propto k^{-3}$. At the Reynolds numbers used here the measured spectrum is much steeper: between $k = 16$ and $k = 63$ the slope is about $-5.5$ for forced turbulence at $Re = 1000$, the same at $256^2$ and $512^2$. Viscosity and drag steepen the cascade, so the scales the network has to reconstruct lie in the steep, near-dissipative tail, not in a clean $k^{-3}$ range.

## 2.2 Numerical method

- **Pseudo-spectral discretisation.** Derivatives are exact in Fourier space, and nonlinear terms are dealiased with the 2/3 rule.
- **Time stepping.** Integrating-factor fourth-order Runge–Kutta: viscosity and drag are integrated exactly, and the step follows a CFL limit of 0.5, rechecked every 10 steps.
- **Resolution.** The DNS runs on $256^2$ or $512^2$ points, depending on regime and $Re$. A run counts as resolved if the enstrophy spectrum at the dealiasing cut-off is below $10^{-3}$ of its peak, and every run records this check.
- **Saved fields.** Snapshots are spectrally truncated to $128^2$ (all modes with $|k| < 64$). $u$, $v$, $p$ and $\omega$ are computed from the vorticity spectrum, with the pressure from $\nabla^2 p = 2(u_x v_y - u_y v_x)$.

Each step takes about 10 ms at $256^2$ and 48 ms at $512^2$ on one core. At $Re = 1000$ the $256^2$ and $512^2$ runs give the same time-averaged spectrum to within about 10% at every saved wavenumber.

## 2.3 Dataset split

Every (regime, $Re$, seed) run starts from its own random field. The split is by whole Reynolds numbers:

Table 2: Reynolds numbers in each split. Each $Re$ has runs in both regimes.

| Split | Reynolds numbers | Seeds |
|---|---|---|
| Training | 250, 400, 1000, 1600, 2500 | 0, 1, 2 |
| Validation (selection only) | 630 | 0, 1, 2 |
| Test, interpolation | 320, 800, 2000 | 10, 11 |
| Test, extrapolation | 4000 | 10, 11 |

The test set holds 560 snapshots. Forced runs at $Re = 4000$ need $512^2$ DNS and take about an hour each; the other runs take minutes.

### 2.4 Coarse input

The coarse input is the $4 \times 4$ block average of each fine field. Block averaging keeps $1/16$ of the Fourier modes, so many fine fields map to the same coarse one, and the network has to supply every scale between $k = 16$ and $k = 63$. The model sees the coarse field bilinearly upsampled back to $128^2$ and predicts a correction on top of it.

Normalisation statistics come from the training split only and are stored in each checkpoint. Training samples are mirrored in $x$ or $y$, with the sign changes the Navier–Stokes equations require, and shifted randomly in the periodic box. Turbulence inputs get no added noise (Section 5).

## 3 Model

### 3.1 Architecture

The network is a U-Net [7] with 7 315 973 parameters:

$$\begin{aligned}
&\text{encoder:} \quad (4, H, W) \to (64, H, W) \to (128, \tfrac{H}{2}, \tfrac{W}{2}) \to (256, \tfrac{H}{4}, \tfrac{W}{4}) \to (512, \tfrac{H}{8}, \tfrac{W}{8}),\\
&\text{decoder:} \quad (512, \tfrac{H}{8}, \tfrac{W}{8}) \to (256, \tfrac{H}{4}, \tfrac{W}{4}) \to (128, \tfrac{H}{2}, \tfrac{W}{2}) \to (64, H, W) \to (4, H, W),
\end{aligned}$$

with skip connections between encoder and decoder at each resolution. Each block is Conv3 $\times$ 3, BatchNorm and ReLU, applied twice. Encoder blocks use dropout ($p = 0.1$), and upsampling is bilinear followed by a $1 \times 1$ convolution.

- **Residual output.** The output is the input plus a final $1 \times 1$ convolution that starts at zero, so an untrained model returns its input.
- **Periodic domains.** Every convolution uses circular padding, and upsampling wraps around. The network is then exactly equivariant to shifts by multiples of 8 cells, which a test checks.
- **Divergence-free projection.** The predicted velocity is projected onto divergence-free fields in Fourier space, $\hat{\mathbf{u}} \leftarrow \hat{\mathbf{u}} - \mathbf{k}(\mathbf{k} \cdot \hat{\mathbf{u}})/|\mathbf{k}|^2$, in physical units. This is an orthogonal projection, so it can never increase the velocity error.
- **Reynolds-number head.** A small regressor predicts $\log_{10} Re$ from the bottleneck features. It is auxiliary and plays no part in checkpoint selection.

## 3.2 Loss

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_{\text{div}}\mathcal{L}_{\text{div}} + \lambda_{\text{vort}}\mathcal{L}_{\text{vort}} + \lambda_{Re}\mathcal{L}_{Re} \left[ + \lambda_{\text{spec}}\mathcal{L}_{\text{spec}} \right], \qquad \lambda_{\text{div}} = \lambda_{\text{vort}} = \lambda_{Re} = 0.01. \tag{2}$$

Table 3: Loss terms. All physics terms are computed in physical units.

| Term | Definition | Purpose |
|---|---|---|
| Reconstruction | mean of $w_c\|\hat{f}_c - f_c\|^2/\|f_c\|^2$ per sample and channel, $w = [3, 3, 2, 2]$ | pointwise fidelity |
| Divergence | mean of $(\partial_x\hat{u} + \partial_y\hat{v})^2$ | incompressibility |
| Vorticity | mean of $\big((\hat{\omega} - (\partial_x\hat{v} - \partial_y\hat{u}))/\sigma_\omega\big)^2$ | $\hat{\omega}$ consistent with $(\hat{u}, \hat{v})$ |
| $Re$ | squared error of normalised $\log_{10} Re$ | $Re$-aware bottleneck |
| Spectral (optional) | mean over $k \geq 16$ of $(\ln E_{\text{pred}}(k) - \ln E_{\text{true}}(k))^2$, energies floored at $10^{-7}$ of the total | fine-scale energy; ignores phase |

The reconstruction loss uses the error relative to each sample's own field, so weak fields count as much as strong ones. Derivatives are exact spectral derivatives. On DNS fields both physics terms are zero to round-off, which a test checks.

## 3.3 Training

Training uses AdamW with a cosine learning-rate schedule from $10^{-3}$ to $10^{-5}$, weight decay $10^{-4}$, batch size 16 and gradient clipping at norm 1. The checkpoint with the lowest validation relative $L_2$ error, averaged over $u$, $v$, $p$ and $\omega$, is kept. The final model (run 03) trained for 300 epochs, and its best checkpoint is at epoch 287.

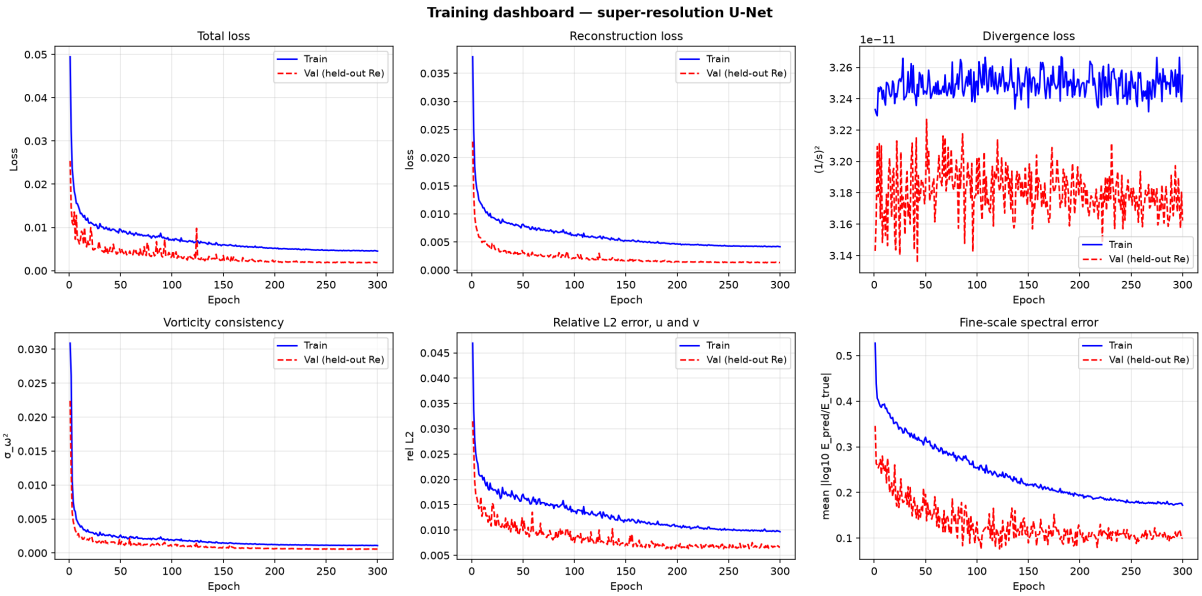



Figure 1: Training curves of run 03. The validation curves (held-out $Re = 630$) lie below the training curves throughout. The velocity error flattens out after about epoch 200. The divergence loss stays at $\sim 3 \times 10^{-11}$ because the projection makes the predicted velocity divergence-free.

# 4 Results

## 4.1 Metrics and baselines

All errors are relative $L_2$ errors in physical units on the test set. The model is compared with three baselines built from the same $32^2$ block averages:

- **bilinear**: the model's own input;
- **bicubic** interpolation;
- **nearest train**: the training snapshot whose coarse field is closest. Its error of about 100% confirms that the test runs share nothing with the training runs.

Two further metrics describe the fine scales. The *spectral error* is

$$\left\langle \left| \log_{10} \left( E_{\mathrm{pred}}(k) / E_{\mathrm{true}}(k) \right) \right| \right\rangle_{k=16,\ldots,63},$$

the mean over the wavenumbers the input cannot represent; shells below $10^{-7}$ of a sample's total energy are ignored. The *HF energy* is the fraction of the true energy at $k \geq 16$ that the prediction recovers.

## 4.2 Reconstruction error

Table 4: Relative $L_2$ error on unseen $Re$, all 560 test snapshots.

| | $u, v$ | | $p$ | | $\omega$ | |
|---|---|---|---|---|---|---|
| | forced | decaying | forced | decaying | forced | decaying |
| **CNN (run 03)** | **0.9%** | **1.3%** | **1.1%** | **1.6%** | **8.4%** | **10.1%** |
| bicubic | 4.4% | 6.1% | 2.5% | 5.0% | 24.1% | 27.0% |
| bilinear (input) | 8.0% | 10.2% | 6.8% | 10.5% | 28.9% | 32.9% |
| nearest train | 95% | 101% | 99% | 105% | 101% | 104% |

Table 5: Error per Reynolds number, velocity / vorticity.

| | $Re = 320$ | 800 | 2000 | 4000 (extrapolation) |
|---|---|---|---|---|
| forced | 0.4% / 2.4% | 0.6% / 5.5% | 1.0% / 10.6% | 1.4% / 14.9% |
| decaying | 0.4% / 2.0% | 0.8% / 6.0% | 1.7% / 13.0% | 2.4% / 19.3% |

The CNN beats bicubic interpolation on every channel and regime: its error is about 5 times lower for velocity and about 3 times lower for vorticity. The error grows smoothly with $Re$, because higher-$Re$ fields carry more energy in the scales the input cannot see, and it shows no jump at the extrapolated $Re = 4000$. Vorticity is the hardest channel because it weights the fine scales by $k^2$.

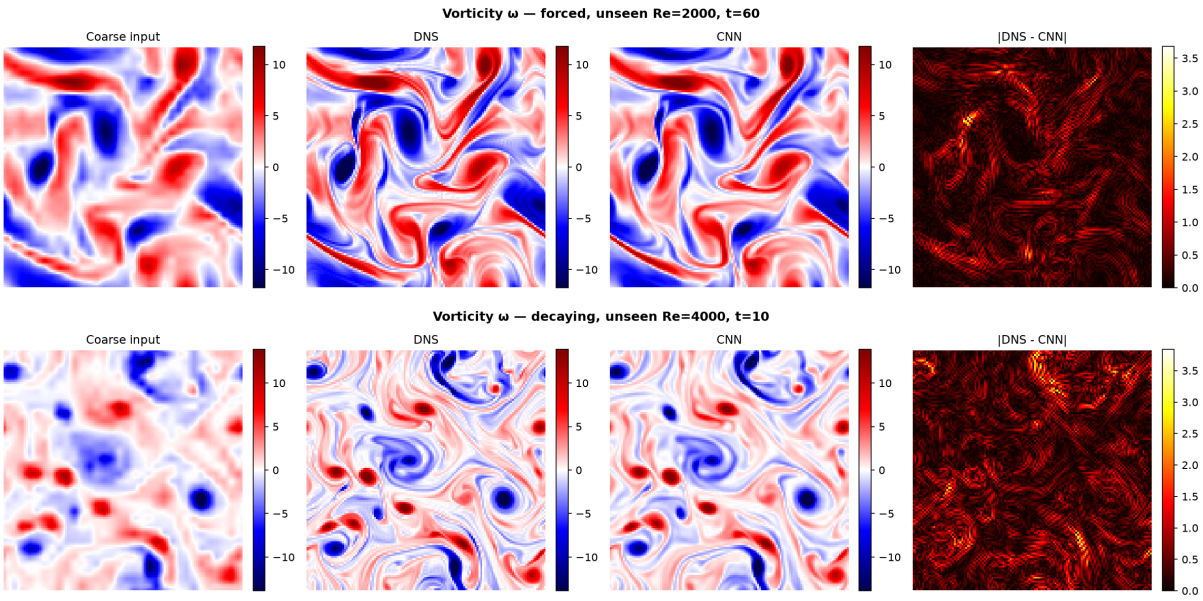
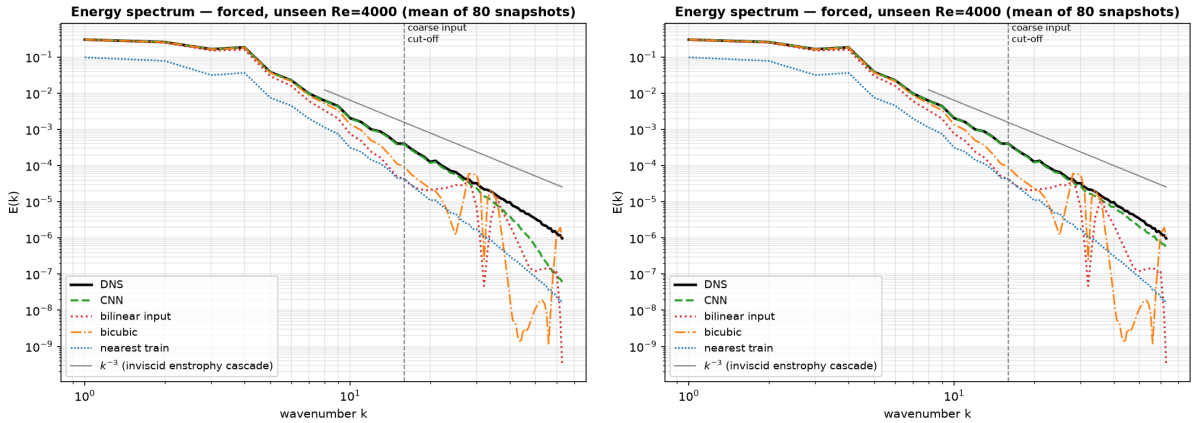



Figure 2: Vorticity at unseen $Re$: coarse input, DNS, CNN prediction and absolute error. Top: forced turbulence at $Re = 2000$. Bottom: decaying turbulence at $Re = 4000$, outside the training range.

The CNN recovers the thin vorticity filaments that the coarse input blurs out. The remaining error is concentrated along the sharpest filaments, where the input says least about their exact position and width.

## 4.3 Fine-scale energy




(a) run 03, $\lambda_{\text{spec}} = 0$ (b) run 04b, $\lambda_{\text{spec}} = 0.002$

Figure 3: Mean energy spectrum of forced turbulence at the extrapolated $Re = 4000$ (80 snapshots), with the coarse cut-off at $k = 16$ and a $k^{-3}$ reference.

Pointwise losses have a known weakness. Part of the fine-scale field cannot be determined from the coarse input, and a pointwise loss replaces that part with its conditional mean, which carries too little energy. In Fig. 3(a) the run 03 spectrum follows the DNS up to about $k = 30$ and falls about $10\times$ below it by $k = 63$. Longer training makes this worse, not better: going from 150 to 300 epochs lowered the HF energy from 94–95% to 91–92% overall, because the model moves further towards the conditional mean.

The optional spectral loss compares $\ln E(k)$ at $k \geq 16$ and ignores phase. It restores the energy (Fig. 3b), but some of the restored structure sits at the wrong phase, and an $L_2$ metric counts misplaced structure twice. Before the spectral runs, three conditions were set for replacing run 03: HF energy of at least 95% at $Re = 2000$ and 4000, a lower spectral error, and a velocity error at most 0.2 points higher with the vorticity error no worse.

Table 6: Pointwise error against fine-scale statistics, test set.

| $\lambda_{\text{spec}}$ | $u, v$ forced / decay. | $\omega$ forced / decay. | $p$ forced / decay. | spec. err forced / decay. | HF energy, Re 2000 / 4000 forced; decaying |
|---|---|---|---|---|---|
| 0 (run 03) | **0.9 / 1.3%** | **8.4 / 10.1%** | 1.1 / 1.6% | 0.226 / 0.247 | 89.4 / 87.0; 89.9 / 87.0% |
| 0.002 (run 04b) | 1.0 / 1.5% | 9.4 / 11.0% | **1.0 / 1.6%** | 0.076 / 0.087 | 95.3 / 93.2; 94.6 / 91.9% |
| 0.01 (run 04a) | 1.1 / 1.6% | 9.7 / 11.3% | 1.2 / 1.8% | **0.066 / 0.061** | **95.5 / 95.3; 95.6 / 94.3%** |

Neither spectral run met all three conditions. Both raise the vorticity error by about

one point, and both stay below 95% HF energy for decaying flow at $Re = 4000$. Run 03 therefore stays the default. The recommendation depends on the use:

- for the reconstructed field itself (pointwise accuracy), use run 03;
- for realistic fine-scale statistics (spectra, small-scale energy), use $\lambda_{\text{spec}} = 0.002$. It keeps most of the spectral gain of $\lambda_{\text{spec}} = 0.01$, with a spectral error 3× below run 03, at a lower pointwise cost.

$\lambda_{\text{spec}} = 0.01$ adds little: 0.002 is better on every pointwise metric and only slightly worse spectrally. Getting realistic fine scales and the lowest pointwise error at once would need a generative model that samples the fine scales, such as diffusion-based super-resolution, instead of predicting a single field.

## 4.4 Reynolds-number estimate and latent space

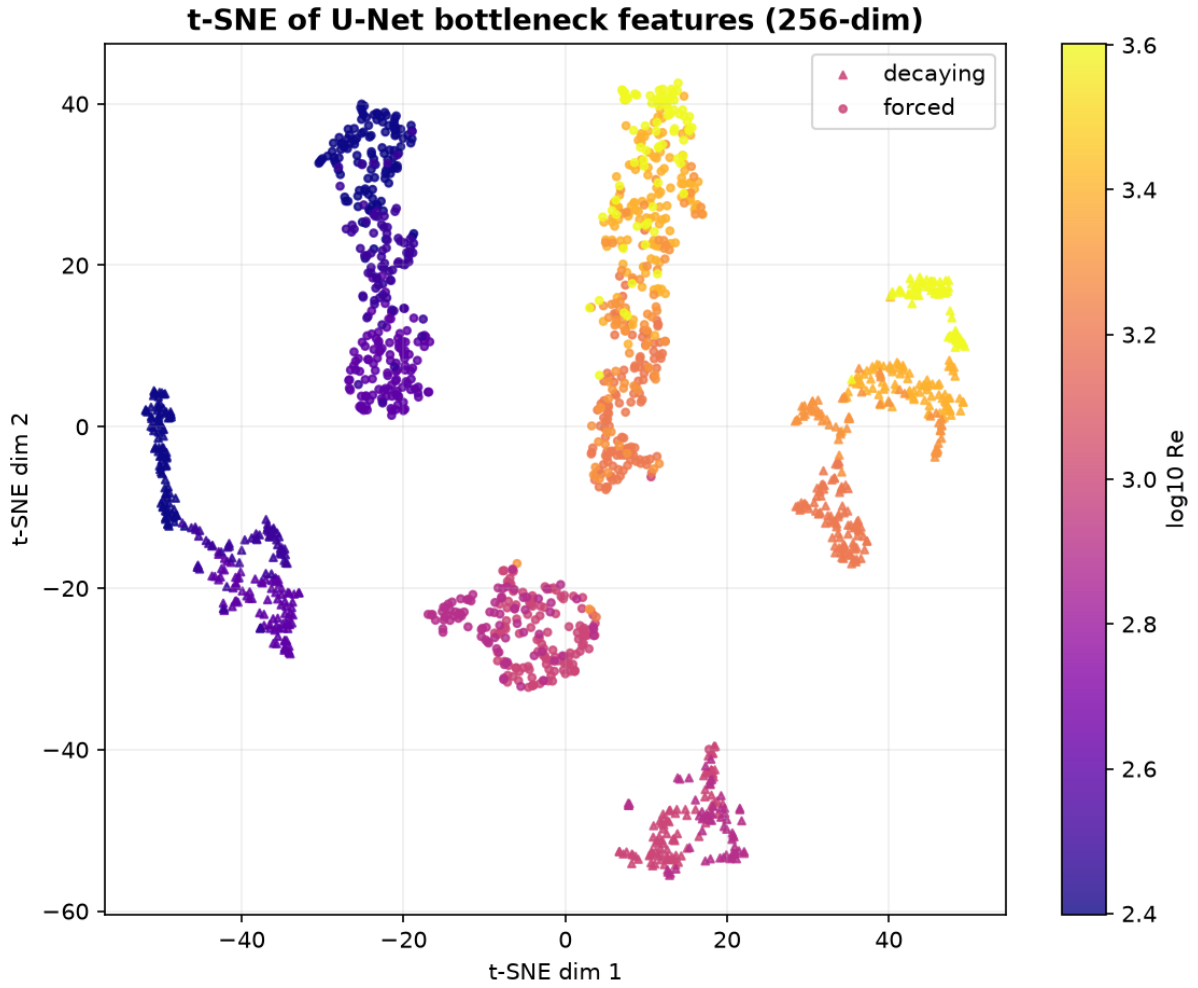



Figure 4: t-SNE of the 256-dimensional bottleneck features of training and test snapshots, coloured by $\log_{10} Re$. Circles: forced; triangles: decaying.

The bottleneck features group by regime and by Reynolds number, with forced and decaying snapshots in separate clusters. The auxiliary $Re$ estimate is still the weakest output: its error on unseen $Re$ is 14% (decaying) and 20% (forced), and 31 to 39% at the extrapolated $Re = 4000$, which it underestimates.

# 5 Experiment Log

**Run 01 (150 epochs).** The first run used a mean-squared-error loss on globally normalised fields, input noise of 0.05, no projection, and checkpoint selection by total validation loss. Its velocity error was 1.7% (forced) and 3.4% (decaying), but its decaying pressure error, 13.7%, was worse than bicubic (5.0%). Checks on that model led to four changes, which are now the turbulence defaults:

- The input noise carried 1.1 to 1.4 times the energy of the true fine scales, so the model learned to add fine scales to clean inputs (validation HF energy up to 154%). Turbulence inputs now get no noise.
- The decaying pressure error correlated ($-0.89$) with how weak the sample's pressure was: a loss on globally normalised fields underweights weak fields. The loss is now relative per sample and channel, in physical units.
- Projecting the predicted velocity onto divergence-free fields cut the mean divergence from 0.09 to about $10^{-6}$ and lowered the velocity error. The projection is now built in.
- The total validation loss swung with the $Re$ error (38 to 150%), so it chose the checkpoint. Selection now uses the relative $L_2$ error averaged over $u$, $v$, $p$ and $\omega$.

**Run 02 (150 epochs).** With the new defaults every number improved. Decaying pressure went from worse than bicubic to 3× better than it, at 1.6%. Validation error was still falling at epoch 150.

**Run 03 (300 epochs).** The final model (Section 4). Doubling the epochs gave small pointwise gains and lowered the fine-scale energy, which motivated the spectral runs 04a and 04b.

# 6 Lid-Driven Cavity Test Case

The same pipeline handles laminar lid-driven cavity flow at $Re = 100$ to 3200, super-resolved from $16 \times 16$ to $64 \times 64$. Each dataset records whether its domain is periodic or walled, and every stage adapts to it: walled data uses central differences on interior nodes

and no divergence-free projection. The cavity solver uses Chorin's projection method [6] on a collocated $127^2$ grid with MUSCL/van Leer advection and a DCT pressure solve, saved at $64^2$. It is validated against the centreline profiles and primary-vortex data of Ghia *et al.* [5]. Above $Re \approx 1000$ the $127^2$ grid under-resolves the flow ($\psi_{\min}$ is 9% off at $Re = 1000$), and `--refine 4` fixes this. The flows are laminar and become steady, so this case tests the walled code path rather than turbulence.

# 7 Limitations

- The Reynolds numbers are limited to a few thousand. Resolved forced DNS above $Re \approx 2500$ needs $512^2$ points or more, and the CPU cost per unit time grows as $N^3$.
- The model is two-dimensional. Three-dimensional turbulence, with its forward energy cascade and $k^{-5/3}$ range, would need a 3D solver and model.
- Forced snapshots are 2 time units apart, so consecutive snapshots of one run are correlated. The split is by $Re$, so this does not leak into the test set.
- The model predicts one field per input. It cannot represent the spread of fine-scale fields consistent with a coarse input.

# 8 Conclusion

A U-Net with residual output, circular padding and a built-in divergence-free projection reconstructs 2D turbulence from 4× coarser inputs at Reynolds numbers it never saw. Its velocity error is 0.9% for forced and 1.3% for decaying flow, 5 times lower than bicubic interpolation, and it degrades smoothly up to the extrapolated $Re = 4000$. With pointwise losses the finest scales come out too weak in energy. A spectral loss fixes the energy but costs pointwise accuracy, so the choice of model depends on whether the reconstructed field or its statistics matter. Sampling the fine scales with a generative model is the natural next step.

# References


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